

A novel bilateral cross-attention network for multi-label fundus disease diagnosis

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HIGHLIGHTS

- Bilateral cross-attention improves multi-label fundus disease diagnosis.
- Cross-attention captures complementary cues from the left and right eyes.
- Gated fusion enables adaptive integration of bilateral eye features.
- Attention-enhanced pooling strengthens discriminative feature learning.
- Grad-CAM shows that the model focuses on disease-related retinal regions.

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ABSTRACT

Multi-label fundus disease classification aims to assign multiple ocular disease labels to bilateral fundus images, enabling automated screening and clinical decision support. In contrast to most existing computer-aided systems that process each eye independently and rely on shallow feature fusion, we explicitly model binocular structure and adaptively combine information from both eyes to handle complex, real-world fundus scenes. To this end, we introduce **DualCrossAttnNet**, a multi-label fundus disease classification network specifically designed for binocular analysis. In particular, we obtain high-resolution bilateral representations using an EfficientNet-B2 backbone and a cross-attention module that jointly reasons about spatial and channel-wise interactions between the left and right eyes. We further employ a gated fusion mechanism to adaptively weight the contributions of each eye, and an SE attention module to recalibrate channel responses before global aggregation. Coupled with a fundus-oriented preprocessing pipeline and a GeM-based classifier, the proposed framework can accurately predict multiple co-occurring ocular diseases from complex clinical fundus images. On the public ODIR-2019 dataset, DualCrossAttnNet substantially improves multi-label classification performance, achieving 88.20% accuracy, 90.98% F1, and 98.49% AUC, with a composite score of 92.73%. These results surpass recent CNN, GNN, and Transformer-based baselines by up to 20.37 percentage points in composite score, demonstrating that DualCrossAttnNet is an effective and scalable solution for intelligent fundus disease diagnosis.

1. Introduction

Fundus diseases have become a primary cause of visual impairment worldwide. According to statistics from the World Health Organization, approximately 2.2 billion people suffer from some form of vision impairment, and nearly half of these cases could be prevented or treated through timely screening and intervention [1]. Fundus imaging, owing to its non-invasive nature, relatively low cost, and rich

informational content, plays a crucial role in diagnosing diseases such as diabetic retinopathy, glaucoma, and age-related macular degeneration. However, traditional manual reading of fundus images is time-consuming, labor-intensive, and heavily dependent on the clinician's experience, which may lead to strong subjectivity and poor inter-observer consistency [2]. With the rapid development of deep learning, AI-based automatic fundus disease recognition methods have

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emerged as powerful tools to support early screening and clinical decision-making [3].

Convolutional neural networks (CNNs) have become the mainstream technique in medical image analysis due to their strong feature extraction capability. Early work mainly focused on binary classification tasks for single diseases, such as the automatic detection or grading of diabetic retinopathy. In real clinical scenarios, however, a single fundus image often exhibits multiple pathological patterns simultaneously, which has driven research interest toward multi-label classification that better reflects clinical needs. For example, Fundus-DeepNet [4] combines high-resolution feature extraction with attention mechanisms, particularly leveraging SE modules to enhance channel responses over lesion regions, thereby improving the accuracy and stability of multi-disease detection. BFENet [5] adopts a two-stream interaction CNN to jointly model bilateral fundus features and verifies the importance of using both eyes. BFPC-Net further introduces attention and feature-fusion modules, enhancing the recognition of key lesion regions and achieving strong performance for small or mixed lesions [6].

Beyond architectural improvements in CNNs, researchers have begun to explore new modeling paradigms. Graph neural networks (GNNs) and Transformer-based architectures offer new approaches for capturing global dependencies. LAD-GCN [7] constructs both a “patient graph” and a “label graph” to explicitly model co-occurrence relationships among disease labels, thereby strengthening contextual reasoning in multi-label classification. MCGL-Net [8] integrates GCNs with LightGBM to achieve multi-scale semantic feature learning. TL-CCL [9] introduces a two-level causal contrastive learning framework that applies causal intervention at both the feature and decision levels, effectively suppressing spurious correlations and improving generalization.

Despite these advances, two key challenges remain. First, bilateral structural information is still underexploited. Many current models are trained and inferred on single-eye images, whereas the anatomical complementarity and pathological consistency between the two eyes are crucial cues in clinical decision-making [2,10,11]. Second, existing feature fusion strategies are relatively simplistic. Many methods rely on direct concatenation or fixed-weight fusion, making it difficult to adaptively modulate the importance of information from different eyes. This can introduce redundant or conflicting features, especially when the quality of the two images is imbalanced, ultimately degrading classification performance [5,7].

Motivated by these issues, we propose DualCrossAttnNet, a bilateral cross-attention network tailored for multi-label fundus disease classification. Specifically, the proposed DualCrossAttnNet takes paired bilateral fundus images as input to fully exploit bilateral structural information. The gated fusion component adaptively balances the contributions from the two eyes to reduce redundancy and mitigate the impact of inter-eye image-quality imbalance.

The main contributions of this work are summarized as follows:

1. We develop a task-specific preprocessing pipeline for fundus image classification to reduce illumination variability and alleviate class imbalance. The pipeline standardizes the input images and enhances lesion saliency through class-aware augmentation, thereby providing more reliable inputs for subsequent bilateral multi-label learning.
2. We develop DualCrossAttnNet, a task-specific bilateral cross-attention network that explicitly models inter-eye structural complementarity and pathological correlations. The adaptive gated fusion mechanism balances the contributions from each eye, leading to a noticeable performance improvement of up to 9.69% over the baseline.
3. Extensive validation on ODIR-2019 (5000 image pairs, eight disease categories) demonstrates 88.20% accuracy, 90.98% F1, and 98.49% AUC, with a composite score of 92.73%. Systematic ablations quantify the synergy among components, and comparative experiments show superiority over representative CNN, GNN, and

Transformer-based methods, supporting the clinical feasibility of the proposed approach for intelligent fundus diagnosis.

2. Related work

Convolutional neural networks (CNNs) have been widely adopted in automated fundus image analysis due to their strong capability for feature extraction and classification. Early studies mainly focused on single-disease detection, while the coexistence of multiple lesions in clinical practice has driven the development of multi-label classification frameworks [12]. Representative approaches, such as Fundus-DeepNet, enhance multi-disease recognition by incorporating attention mechanisms (e.g., SE blocks) to recalibrate channel responses over salient lesion regions, achieving high F1 scores and Kappa coefficients on ODIR-2019 [4]. Subsequent models, including ML-CNN and BFPC-Net, further improve performance through multi-scale convolution and explicit feature fusion, demonstrating the effectiveness of attention-based feature enhancement, particularly for small or mixed lesions [6]. However, the inherently local receptive fields of CNNs limit their ability to model long-range dependencies and complex semantic correlations among labels, which motivates recent explorations of graph-based and Transformer-based architectures.

Beyond CNN-based methods, graph neural networks (GNNs) and Transformer architectures have been explored to model complex lesion interactions and global dependencies in fundus image analysis. Representative GNN approaches, such as LAD-GCN and MCGL-Net, construct patient-label graphs or integrate mixed graph convolution modules to enhance cross-sample and cross-label information propagation, achieving improved F1 and AUC performance on ODIR-2019 [7,8]. However, graph construction and iterative message passing introduce substantial computational overhead and scalability issues, and GNNs are prone to overfitting in small-sample settings with sparse or unreliable similarity structures. Transformer-based methods leverage self-attention to capture long-range dependencies and spatial consistency, with TL-CCL further improving robustness through causal contrastive learning [7]. Despite these advantages, Transformers suffer from high computational and memory costs due to quadratic self-attention, require large-scale training data, and often struggle to preserve fine-grained local lesion details on small medical datasets.

In clinical practice, ophthalmologists routinely examine both eyes to improve diagnostic reliability by leveraging strong anatomical symmetry and correlated or complementary disease manifestations. Motivated by this, bilateral feature fusion has become an important direction in fundus disease classification. Representative methods, such as BFENet, employ dual-stream CNNs to extract features from left- and right-eye images and perform binocular analysis through high-level fusion, achieving notable gains in F1 and AUC over single-eye baselines [5]. Later studies, including BFPC-Net and multi-scale bilateral fusion models, further incorporate attention mechanisms to adaptively reweight bilateral features and enhance robustness under complex lesions and variable imaging conditions [6]. Despite their effectiveness, existing bilateral fusion strategies remain largely shallow, relying on late concatenation or fixed weighting, and are insufficiently adaptive to large inter-eye differences in image quality or lesion severity, which can introduce redundant or conflicting information.

To overcome these limitations, we propose DualCrossAttnNet, a bilateral learning framework that explicitly models deep cross-eye interactions and adaptive information selection. By integrating cross-attention across spatial and channel dimensions, the proposed model enables fine-grained and dynamic feature exchange between the two eyes, allowing salient and complementary cues to be emphasized. Furthermore, a gated fusion mechanism adaptively regulates each eye’s contribution according to image quality and lesion patterns, effectively mitigating redundancy and imbalance. Together with SE-based channel recalibration and GeM pooling-based classifier, DualCrossAttnNet fully exploits binocular structural complementarity and pathological consistency,

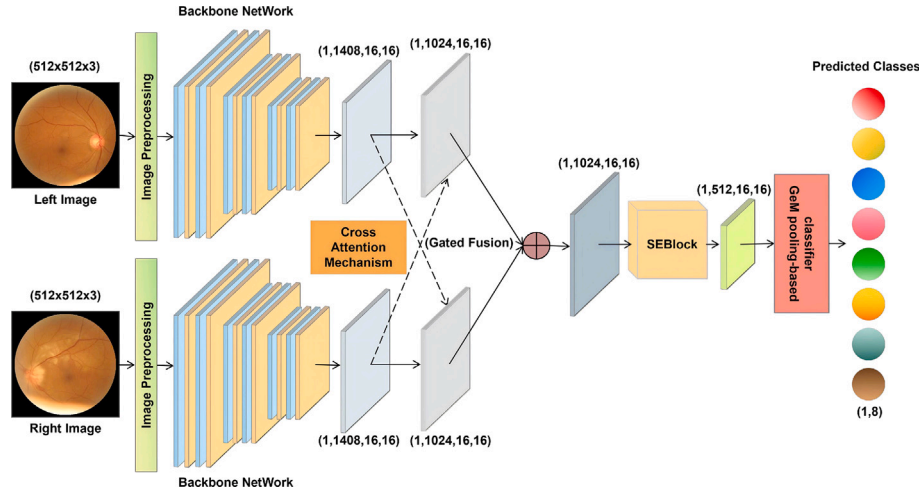


Fig. 1. General diagram of the DualCrossAttnNet.

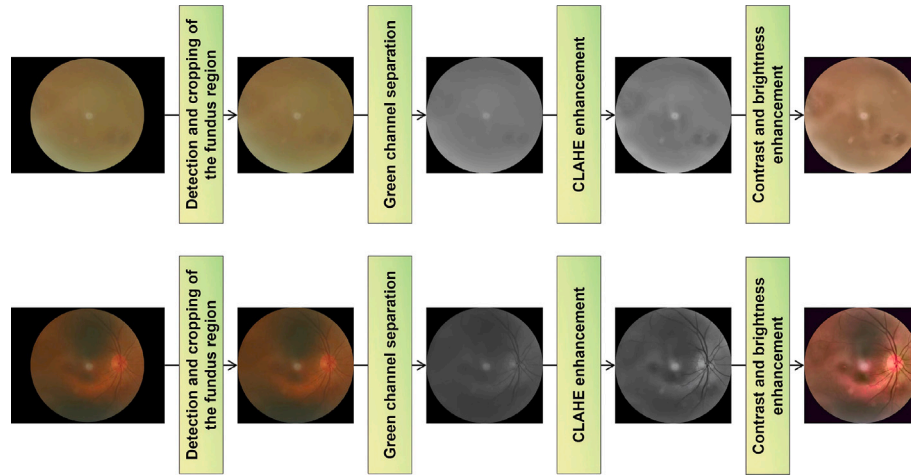


Fig. 2. The preprocessing flowchart of the DualCrossAttnNet.

providing a robust and clinically feasible solution for intelligent fundus diagnosis.

3. Methods

Fig. 1 illustrates the architecture of DualCrossAttnNet, which is designed to perform multi-label classification of bilateral fundus images through the collaborative integration of five functional modules. The image preprocessing module in Fig. 2 standardizes input quality by performing automatic detection, contrast enhancement, and adaptive augmentation. The backbone network in this paper employs a shared-weight EfficientNet-B2 backbone to capture deep visual features from both eyes. The cross-attention mechanism facilitates deep interaction and mutual enhancement of bilateral features across spatial and channel dimensions. The gated fusion module adaptively balances the pixel-level contributions from each eye based on learned quality assessments. Moreover, an SE block is applied for channel recalibration and dimensionality reduction. Finally, a GeM pooling-based classifier is utilized to aggregate global information and generate final probabilities for the eight disease categories. This multi-level attention design explicitly models inter-eye correlations while maintaining computational efficiency.

3.1. Image preprocessing

The fundus images in the ODIR dataset exhibit significant variations in illumination, resolution, and contrast, requiring a standardized preprocessing pipeline to ensure data quality. We designed a four-stage preprocessing pipeline in Fig. 2, which includes eye region detection, image quality enhancement, size normalization, and intelligent data augmentation.

Firstly, a thresholding method is used to detect non-black regions, generating a binary mask by setting pixel values greater than a tolerance threshold (set to 10). The valid fundus region is then extracted and expanded into a square to preserve its geometric properties. Secondly, the green channel undergoes CLAHE processing [13] combined with LAB color space transformation, with CLAHE parameters set to a clip limit of 3.0 and a tile size of 8×8 . This effectively enhances local contrast while suppressing noise amplification. Thirdly, the images are then resized to a standardized size of 512×512 using an interpolation algorithm to ensure high-quality resampling. As a result, an intelligent data augmentation strategy [14] is implemented to eliminate the class imbalance, which includes geometric transformations (rotation, flipping), color adjustments (brightness, saturation, and hue variations), and noise addition (Gaussian noise, motion blur). The aug-

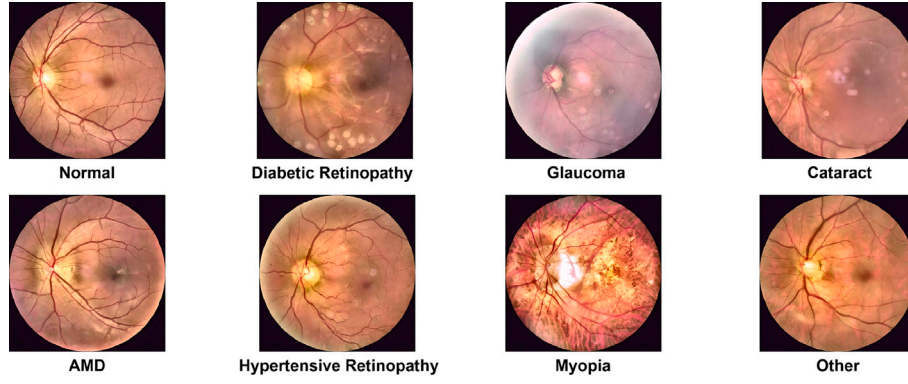


Fig. 3. The result images produced by the preprocessing of the DualCrossAttnNet.

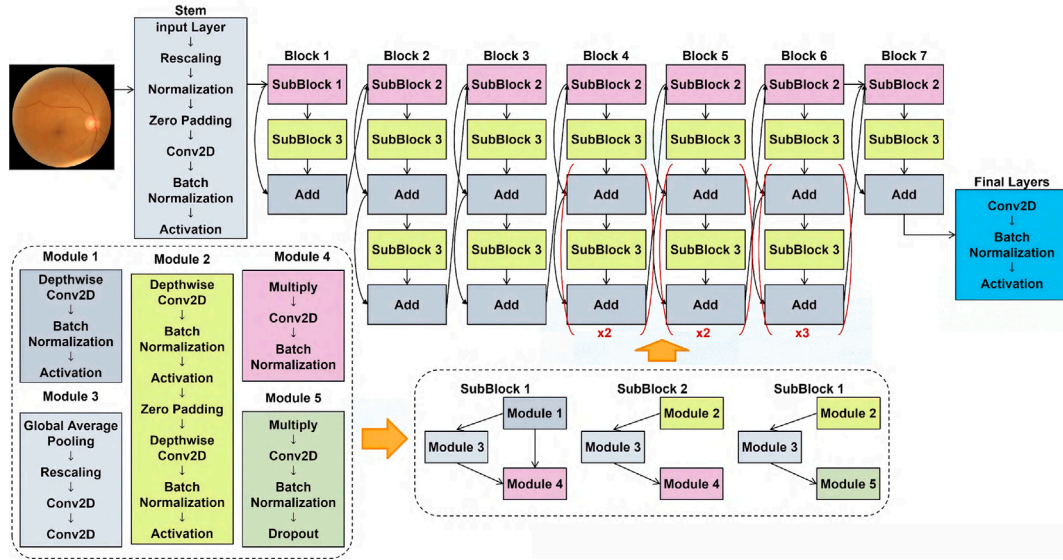


Fig. 4. The backbone network of EfficientNet-B2.

mentation intensity is dynamically adjusted based on the number of samples in each class to ensure a more balanced training data distribution. To reduce the risk of unintended bias, all augmentation operations are applied only to the training split, while the validation and test sets remain unaltered. The augmentation policy is restricted to clinically plausible transformations, including bounded rotation and flipping, mild brightness/saturation/hue adjustment, and low-intensity Gaussian-noise or motion-blur perturbation. These operations are used to improve robustness and class balance without introducing unrealistic anatomical distortion or changing the underlying disease semantics. The preprocessed images from Fig. 3 exhibit the following characteristics:

- **Size normalization:** All images are resized to 512×512 pixels to ensure consistency in the input data.
- **Precise eye region localization:** Through thresholding and a minimum bounding rectangle strategy, black borders and irrelevant backgrounds are effectively removed.
- **Image quality enhancement:** CLAHE and LAB color space processing significantly improve vascular contrast and overall clarity.
- **Feature prominence:** Lesion features become more pronounced, aiding the model's feature learning and disease identification.

3.2. Backbone network

We adopt EfficientNet-B2 [15], shown conceptually in Fig. 4, as the backbone for feature extraction, since it offers a favorable trade-off between computational efficiency and representational capacity. EfficientNet-B2 scales network depth, width, and resolution jointly via compound scaling, achieving strong performance at a relatively low computational cost.

The backbone uses mobile inverted bottleneck convolution (MBConv) blocks with integrated SE attention [16], which enable dynamic recalibration of channel responses. For bilateral processing, we employ a weight-shared dual-branch architecture: the left and right fundus images are passed through identical EfficientNet-B2 networks, ensuring consistent feature extraction. The resulting feature maps have a size of $16 \times 16 \times 1408$ and provide rich spatial and channel information for downstream cross-attention and fusion.

We initialize the backbone with ImageNet-pretrained weights and then fine-tune it on the fundus dataset, which accelerates convergence and improves final performance.

3.3. Cross-attention mechanism

The cross-attention mechanism is the core innovation of DualCrossAttnNet, designed to enable deep interaction and mu-

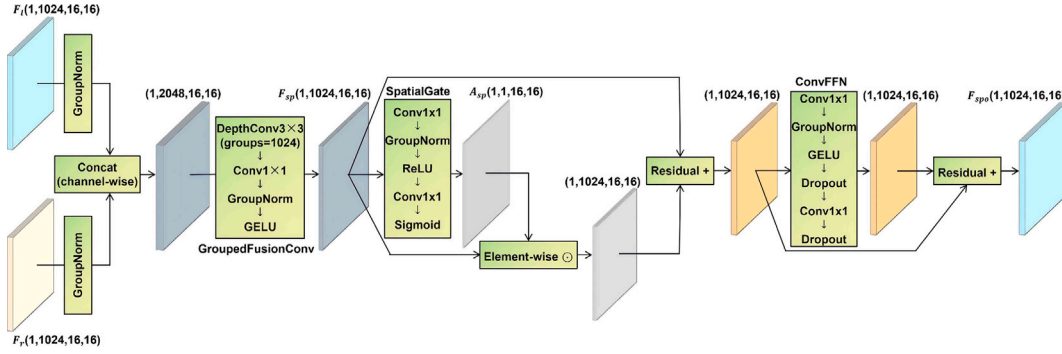


Fig. 5. The spatial attention network of the DualCrossAttnNet.

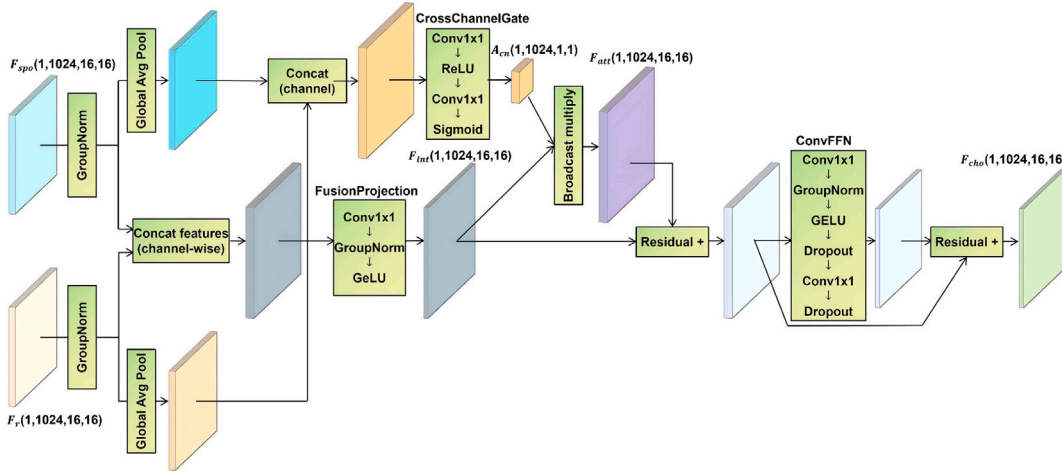


Fig. 6. The channel attention network of the DualCrossAttnNet.

tual reinforcement between left- and right-eye features. It adopts a multi-head attention architecture [17] and uses an input projection layer to reduce the 1408-dimensional features extracted by EfficientNet-B2 to 1024 dimensions, maintaining representational capacity while improving computational efficiency. The mechanism follows a serial design: spatial attention is applied first, and its output is then fed into the channel-attention branch. Each branch comprises two processing layers, and training stability as well as gradient flow are enhanced via DropPath regularization [18] and residual connections [19].

For the spatial attention depicted in Fig. 5, we employ depthwise separable convolutions [20] together with GroupNorm normalization [21] to model dependencies across spatial locations through feature concatenation and multi-layer convolutional operations. Specifically, the input features are first normalized with GroupNorm; the left- and right-eye features are then concatenated along the channel dimension and passed through depthwise separable convolutions (depthwise followed by pointwise) to extract spatial interaction cues. The resulting features are again normalized and activated with GELU [22] to form spatial representations. Spatial attention weights are produced by a dimensionality-reduction convolution, followed by GroupNorm, ReLU, and Sigmoid activation, and are finally applied element-wise to the spatial features to obtain the enhanced spatial representation.

The channel-attention stage shown in Fig. 6 employs global average pooling to obtain channel-wise statistics and a channel-interaction module to learn inter-channel dependencies. Concretely, both the output of the spatial-attention branch and the original features are first normalized with GroupNorm. Global average pooling then compresses spatial information into channel descriptors, which are concatenated across the left

and right eyes and passed through a multilayer perceptron to produce channel-attention weights. In parallel, the normalized features are fused via the channel-interaction module. The resulting attention weights are applied (element-wise, with broadcasting) to the fused features to yield a channel-enhanced representation.

The feature fusion mechanism is computed as Eq (1), where F_{spo} and F_{cho} denote the spatial and channel features, respectively. The learnable parameter α regulates their relative contributions, and the sigmoid function $\sigma(\cdot)$ constrains the weight to the range [0, 1], enabling adaptive fusion of spatial and channel information.

$$F_{en} = \sigma(\alpha) \cdot F_{spo} + (1 - \sigma(\alpha)) \cdot F_{cho}. \quad (1)$$

A key advantage of the proposed cross-attention mechanism is that all computations are performed directly on four-dimensional feature maps $[B, C, H, W]$, thereby avoiding the flattening and reshaping overhead typical of Transformer-style models. Training stability is further improved by the serial processing scheme and residual connections. After each layer, a feed-forward network—comprising two convolutional layers, GELU activation, and Dropout regularization—is applied to transform features, enhancing the model's nonlinear representational capacity.

3.4. Gated feature fusion

The gated fusion module learns adaptive fusion weights to balance the contributions of the left and right eyes. Compared with standard attention/gating fusion blocks, the proposed gated feature fusion is performed after bilateral cross-attention, so that the fusion weights are estimated from cross-eye contextualized features rather than from two

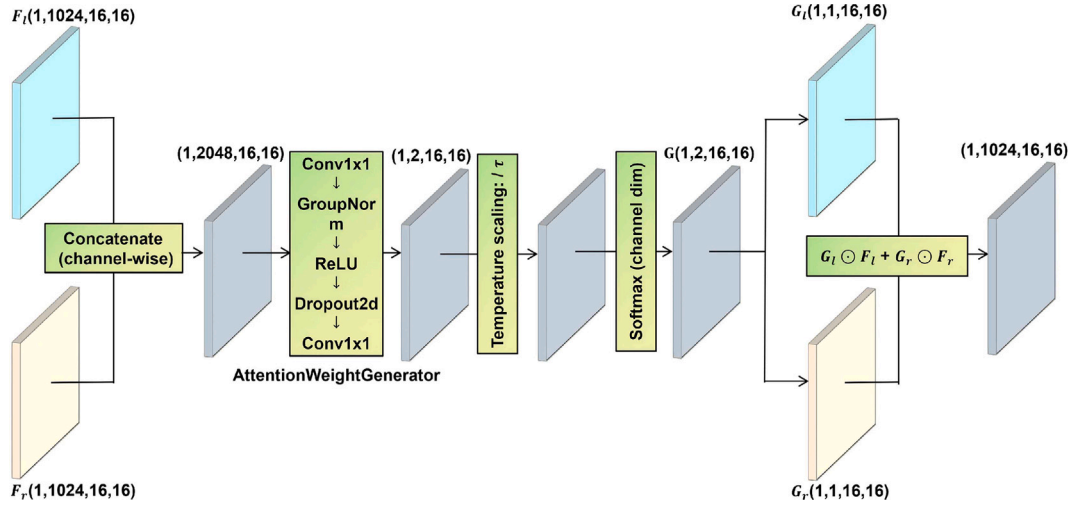


Fig. 7. The gated feature fusion strategy of the DualCrossAttnNet.

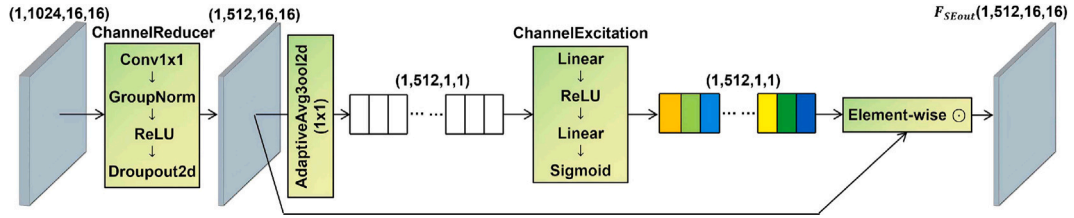


Fig. 8. Squeeze-and-Excitation structure.

isolated branches. In addition, the fusion weights are spatially adaptive instead of globally fixed, which allows the model to modulate the contribution of each eye across different retinal regions. This design improves the flexibility of bilateral feature aggregation under asymmetric lesion distribution and inter-eye image-quality imbalance. As shown in Fig. 7, we first concatenate the cross-attention-enhanced feature maps from both eyes and input them into a compact gated convolutional network.

The gated network consists of two 1×1 convolutional layers. The first layer reduces the channel size from 2048 to 256, while the second layer outputs two channel gating logits. A learnable temperature parameter τ (initialized to 2.0) is used to control the sharpness of the gating distribution. Softmax is then applied along the channel axis to generate normalized fusion weights for the two eyes. The final bilateral features are obtained by a weighted sum of the feature maps, with pixel-wise, spatially varying weights.

To improve stability and generalization, we use GroupNorm instead of BatchNorm for small-batch training, apply 2D Dropout [23] (dropout rate of 0.1) to reduce overfitting, and utilize the temperature parameter to prevent excessive reliance on a single eye. The spatially adaptive weights enable pixel-level fusion, allowing the model to focus more on the eye with higher-quality or more informative lesion patterns at each spatial location.

3.5. SE-based attention enhancement

After gated fusion, the SE (Squeeze-and-Excitation) attention mechanism [16] is applied to further calibrate and enhance the fused features while also performing dimensionality reduction. The SE-based enhancement module in Fig. 8 consists of a channel reduction layer followed by a squeeze-excitation block. First, the feature dimensionality is reduced from 1024 to 512 by a 1×1 convolution, GroupNorm normalization, ReLU activation, and 2D Dropout regularization (dropout rate 0.2). On top of the reduced features, the SE attention module is applied: global

context is captured by compressing the spatial dimensions to 1×1 via global average pooling (AdaptiveAvgPool2d(1)), and a two-layer fully connected network is then used to learn channel-wise weights. Specifically, the first fully connected layer reduces the dimensionality with a reduction ratio of 16 ($512 \rightarrow 32$) and uses ReLU activation, while the second layer restores the channel dimension ($32 \rightarrow 512$) and applies a Sigmoid function to produce weights in the range $[0, 1]$. These learned weights are finally applied to the reduced feature maps, enabling channel-wise recalibration of the fused representation.

Let z denote the channel descriptor obtained after global average pooling and w the excitation weights. The SE operation can be summarized as Eq (2), where W_1 and W_2 are learnable weight matrices, $\delta(\cdot)$ is the ReLU activation, $\sigma(\cdot)$ is the Sigmoid function, $\odot$ denotes element-wise multiplication, and F and $\tilde{F}$ are the input and recalibrated feature maps, respectively. This module not only performs dimensionality reduction ($1024 \rightarrow 512$) but also highlights important feature channels through channel attention. In fundus disease diagnosis, this helps the network focus on feature channels that are most relevant to the diseases.

$$w = \sigma(W_2 \delta(W_1 z)), \quad \tilde{F} = w \odot F. \quad (2)$$

3.6. GeM pooling-based classifier

The final classification head shown in Fig. 9 is built upon generalized mean pooling (GeM) [24]. The GeM-based classifier first applies an attention enhancement module to modulate spatial responses and then aggregates features via GeM with a learnable pooling exponent p . A three-layer fully connected network is used to predict the eight disease labels.

Attention enhancement. A two-layer 1×1 convolutional subnetwork is used. The first layer reduces the channel dimension from 512 to 64 (i.e., $512/8$), followed by GroupNorm (32 groups), ReLU, and 2D Dropout (rate 0.15). The second layer outputs a single-channel attention

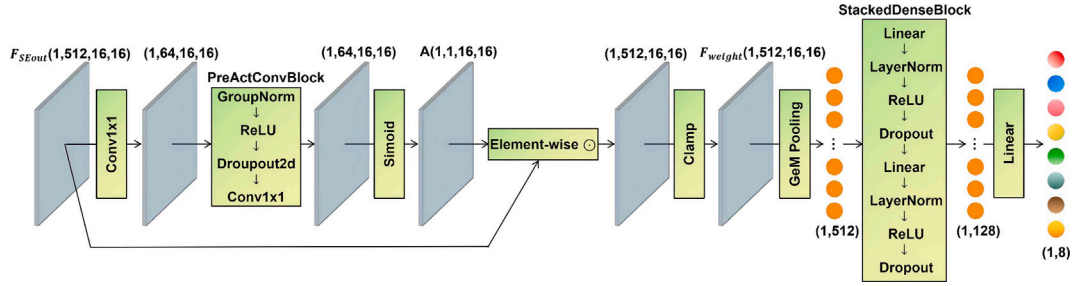


Fig. 9. The GeM pooling-based classifier structure.

map, which is squashed by a Sigmoid function to produce weights in $[0, 1]$. This map is multiplied with the input feature map to generate attention-weighted features.

GeM pooling. GeM aggregates spatial features with a learnable parameter p , initialized to 2.0. Given a feature map $F \in \mathbb{R}^{C \times H \times W}$, GeM pooling is defined in Eq. (3), where c indexes the channel. When $p = 1$, GeM reduces to average pooling; as $p \rightarrow \infty$, it approximates max pooling. This flexibility allows the model to automatically select an appropriate pooling behavior based on the data characteristics.

$$\text{GeM}(F)_c = \left(\frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W |F_{c,ij}|^p \right)^{1/p}, \quad (3)$$

Classification network. The resulting 512-dimensional feature vector is passed through a three-layer multilayer perceptron (MLP) with progressive dimensionality reduction of $512 \rightarrow 256 \rightarrow 128 \rightarrow 8$. Each hidden layer uses LayerNorm [25], ReLU, and Dropout (rates 0.6 and 0.4 for the first and second layers, respectively). The final 8-dimensional logits correspond to the eight ocular disease categories and are used for multi-label prediction via Sigmoid activation during inference.

4. Experimental results

4.1. Experimental settings

This study implements the DualCrossAttnNet system in Python 3.10. Experiments were conducted on a high-performance workstation running Ubuntu 22.04 LTS, equipped with an NVIDIA RTX 4090 GPU (24 GB VRAM), an Intel Core i9-13900K CPU, and 32 GB DDR5 memory, providing sufficient computational resources for deep learning training. The deep learning framework is PyTorch 2.6.0 with CUDA 11.8 for efficient GPU acceleration.

Model training uses the AdamW optimizer [26] with an initial learning rate of 1×10^{-4} , batch size 32, weight decay 0.01, and a maximum of 100 epochs. The loss function is BCEWithLogitsLoss for multi-label classification. A comprehensive monitoring suite tracks loss, accuracy, precision, recall, F1, AUC, and Kolmogorov–Smirnov (KS) statistics during training and validation. To mitigate overfitting, we adopt a ReduceLROnPlateau scheduling strategy: if the validation loss does not improve for 15 consecutive epochs, the learning rate is halved (factor 0.5) with a minimum of 1×10^{-7} ; an early-stopping criterion terminates training when the validation F1 shows no improvement for 10 consecutive epochs. We employ mixed-precision training [27] and gradient clipping (max norm 1.0) [28] to stabilize optimization, and fix the random seed (42) to ensure reproducibility.

4.2. Dataset

We conduct experimental validation on the ODIR-2019 (Ocular Disease Intelligent Recognition) dataset [29]. The dataset originates from the International Fundus Image Intelligent Recognition Competition organized by Peking University. It was curated by the National Institute of Health and Medical Data at Peking University with

Table 1

Disease category distribution statistics of the ODIR dataset. Normal (N), Diabetes (D), glaucoma (G), Cataract (C), Age-related Macular Degeneration (A), Hypertension (H), pathological Myopia (M), and other diseases/abnormalities (O).

Label	(N)	(D)	(G)	(C)	(A)	(H)	(M)	(O)
Sample size	1626	1618	305	308	238	149	243	1390

technical support from the Institute of Artificial Intelligence. As one of the most authoritative publicly available multi-label fundus datasets, ODIR-2019 contains 5000 pairs of de-identified bilateral color fundus images (10000 images in total), with each pair accompanied by clinical metadata. The dataset covers eight major categories shown in Table 1.

As shown in Fig. 10, all images in the dataset are sourced from real clinical environments at collaborating hospitals and medical institutions and are annotated using a multi-label approach, where a single image may be labeled with multiple diseases. All diagnostic keywords are provided by experienced ophthalmologists and undergo a strict quality-control process to ensure both accuracy and consistency.

The dataset is divided into training, validation, and test sets in an 80%, 10% and 10% ratio. We split the dataset into 4000 image pairs (8000 images) for training, 500 image pairs (1000 images) for validation, and 500 image pairs (1000 images) for testing. The detailed distribution of each ophthalmic category in this dataset is presented in Table 2. To ensure a scientifically sound partitioning of the data, a stratified random sampling strategy is used, with splitting performed at the patient level. This ensures that both eyes of the same patient are always assigned to the same subset, avoiding data leakage. Additionally, to accommodate the multi-label nature of the data, the distribution of each disease category in the training, validation, and test sets is kept consistent, effectively preventing model bias caused by imbalanced data distribution. A fixed random seed (seed = 42) is used to ensure reproducibility.

4.3. Evaluation metrics

To comprehensively evaluate the performance of multi-label fundus disease classification, we compute seven complementary evaluation metrics, including Accuracy, Precision, Recall, F1-score, Area under the receiver operating characteristic curve (AUC-ROC), the Kolmogorov–Smirnov (KS) statistic and composite score. Accuracy reflects the overall proportion of correctly predicted labels. Precision measures the reliability of positive predictions, while Recall evaluates the model's ability to identify true positive cases. The F1-score, provides a balanced indicator of detection performance under class imbalance. The Area under the receiver operating characteristic curve (AUC-ROC) quantifies the model's discriminative capacity across varying decision thresholds. Additionally, the Kolmogorov–Smirnov (KS) statistic captures the maximum separation between the predicted probability distributions of positive and negative samples, offering a sensitive

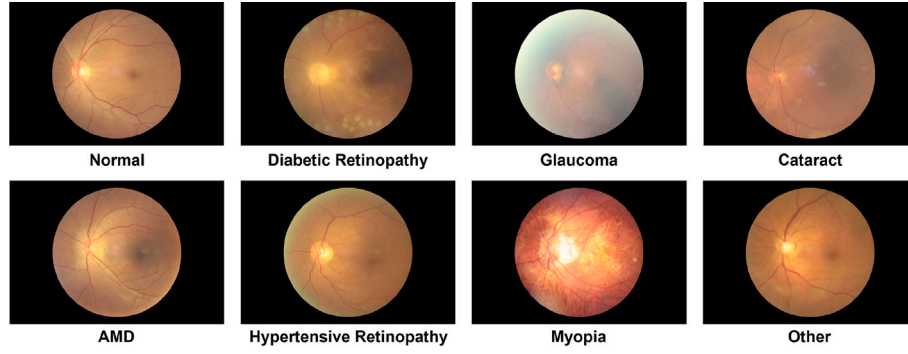


Fig. 10. Example of original bilateral fundus images in the ODIR dataset.

Table 2

Dataset distribution across training, validation, and test sets.

Label	N	D	G	C	A	H	M	O
Training set	1295	1307	249	239	192	113	188	1119
Validation set	156	160	38	29	25	13	32	135
Test set	175	151	18	40	21	23	23	136

measure of classification separability. We define a composite score in Eq. (4) as a weighted average of the above metrics (Accuracy, F1, AUC, and KS), designed to reflect overall performance in the multi-label setting. This aggregate indicator is used to compare different model configurations and to quantify the net gain brought by architectural and algorithmic improvements.

To comprehensively evaluate the performance of multi-label fundus disease classification, each patient pair is represented as an 8-dimensional binary target vector corresponding to the eight diagnostic categories, and the network outputs eight logits that are converted into per-label probabilities through a sigmoid function. During inference, these probabilities are binarized using a fixed decision threshold of 0.5. Unless otherwise stated, Precision, Recall, F1-score, AUC-ROC, and the Kolmogorov–Smirnov (KS) statistic are reported as macro-averaged one-vs-rest multi-label metrics, so that each label is evaluated independently against the remaining labels before aggregate reporting.

The evaluation further includes Accuracy, which reflects the overall proportion of correctly predicted labels. Precision measures the reliability of positive predictions, Recall evaluates the ability to identify true positive cases, and F1-score provides a balanced indicator under class imbalance. AUC-ROC quantifies discriminative capacity across varying decision thresholds, while KS measures the maximum separation between the predicted probability distributions of positive and negative samples. In addition, the composite score in Eq. (4) defines a composite score. It is introduced as a supplementary summary index to facilitate a more intuitive overall comparison across multiple evaluation metrics. The current weight combination was selected empirically because it gives performance-ranking results that are generally consistent with the comparative trends observed from the seven individual metrics.

$$\text{Composite Score} = 0.3 \times \text{Accuracy} + 0.25 \times \text{F1} + 0.25 \times \text{AUC} + 0.2 \times \text{KS}. \quad (4)$$

4.4. Five-fold stratified cross-validation

To further validate the robustness and reproducibility of DualCrossAttnNet under the ODIR-2019 setting, we additionally conducted a five-fold stratified cross-validation study. Considering the multi-label nature of the dataset, we adopted a patient-level MultilabelStratifiedKFold strategy to preserve the label distribution across folds while preventing information leakage between the left and right eyes of the same subject. For each fold, the training process followed the same optimization settings as the main experiment, and the best-performing checkpoint was selected for subsequent evaluation on the independent test set.

The cross-validation results in Table 3 indicate that DualCrossAttnNet is stable across different data partitions. The mean composite score is 92.14% with a standard deviation of 0.81%, suggesting limited performance fluctuation among folds. In particular, the standard deviation of AUC is 0.54%, indicating highly consistent ranking ability, while the standard deviation of KS is 0.58%, further confirming the stability of class separability. The F1-score remains within a relatively narrow range across folds, whereas Recall shows slightly larger variation, which may be associated with differences in the distribution of minority classes among folds. Overall, the five-fold cross-validation results are highly consistent with the original single-split evaluation, thereby providing stronger evidence for the reliability and reproducibility of the proposed framework.

4.5. Comparison of backbone networks and preprocessing

The choice of backbone network has a decisive impact on fundus disease classification performance. We therefore compare several mainstream CNN backbones initialized with ImageNet [30] pretrained weights and fine-tuned on ODIR-2019 under a unified training protocol. For each backbone, we conduct experiments both with and without the proposed fundus-oriented preprocessing pipeline to quantify the benefits of preprocessing.

Tables 4 and 5 report the performance on the validation and test sets with and without preprocessing, respectively. The results show that incorporating the preprocessing pipeline consistently improves accuracy, F1-score, AUC, KS, and the composite score for all backbone networks.

Table 3

Five-fold stratified cross-validation results of DualCrossAttnNet on ODIR-2019.

Fold	Accuracy	Precision	Recall	F1	AUC	KS	Composite
Fold 1	87.00	90.16	86.34	88.17	96.80	93.65	91.07
Fold 2	88.40	91.87	88.16	89.94	97.92	94.88	92.46
Fold 3	87.60	90.70	89.00	89.76	97.39	94.02	91.87
Fold 4	89.00	93.02	90.44	91.68	98.42	95.27	93.28
Fold 5	87.80	91.11	88.07	89.51	97.72	94.41	92.03
Mean $\pm$ Std	87.96 $\pm$ 0.69	91.37 $\pm$ 1.00	88.40 $\pm$ 1.34	89.81 $\pm$ 1.12	97.65 $\pm$ 0.54	94.45 $\pm$ 0.58	92.14 $\pm$ 0.81

Table 4
Performance comparison of backbone networks with preprocessing data.

Test set							
Model	Accuracy	Precision	Recall	F1	AUC	KS	Score
ResNet18	61.25	58.12	52.43	55.16	78.88	51.91	62.27
ResNet34	62.18	59.01	53.03	56.78	78.04	49.81	62.08
ResNet50	65.89	62.25	58.23	60.87	80.00	56.12	66.74
ResNet101	64.23	61.34	54.36	57.64	77.89	41.92	61.54
EfficientNet-B2	73.45	72.18	69.15	70.20	81.34	62.18	72.36
EfficientNet-B3	68.09	74.01	66.33	69.96	87.82	61.73	72.22
Validation set							
Model	Accuracy	Precision	Recall	F1	AUC	KS	Score
ResNet18	62.85	58.38	53.74	55.96	81.81	35.36	64.17
ResNet34	63.75	60.96	54.62	56.27	82.02	56.12	66.12
ResNet50	67.37	64.18	59.82	61.92	82.78	58.43	68.76
ResNet101	65.65	62.97	56.07	59.32	79.78	45.84	63.64
EfficientNet-B2	75.12	72.39	70.64	71.75	86.27	64.93	74.13
EfficientNet-B3	69.15	75.57	66.56	71.34	90.21	64.24	74.17

Table 5
Performance comparison of backbone networks without preprocessing data.

Test set							
Model	Accuracy	Precision	Recall	F1	AUC	KS	Score
ResNet18	60.89	57.45	52.18	54.69	78.12	50.65	61.64
ResNet34	61.78	58.34	52.95	55.31	78.67	49.12	61.62
ResNet50	62.23	58.67	54.46	56.48	79.89	50.34	62.83
ResNet101	60.12	56.89	51.78	54.21	75.34	46.78	59.78
EfficientNet-B2	67.23	66.78	63.45	65.07	84.12	58.67	69.20
EfficientNet-B3	62.45	68.23	61.34	64.60	81.23	56.78	66.55
Validation set							
Model	Accuracy	Precision	Recall	F1	AUC	KS	Score
ResNet18	61.73	58.92	54.28	56.50	80.95	73.32	63.64
ResNet34	62.85	60.18	55.34	56.36	81.53	52.15	64.01
ResNet50	63.45	60.23	55.89	57.98	81.67	52.32	64.39
ResNet101	63.23	58.45	53.67	55.96	77.89	48.34	61.30
EfficientNet-B2	69.12	65.45	66.90	68.55	85.39	61.34	71.18
EfficientNet-B3	64.23	69.78	62.45	65.91	84.35	59.12	68.43

By enhancing local contrast, suppressing illumination variation, and removing irrelevant background regions, preprocessing simplifies the learning problem and promotes the extraction of stable, lesion-relevant features.

We also observe that shallow ResNets (ResNet-18 and ResNet-34) exhibit limited representational capacity and underperform on several metrics, while very deep or wide models increase computational cost without proportionate gains in performance. In contrast, EfficientNet-B2 achieves a favorable balance among network depth, width, and input resolution, yielding strong performance and good parameter efficiency. Based on these observations, we select EfficientNet-B2 as the backbone network for DualCrossAttnNet and adopt the proposed image preprocessing pipeline as an essential component of the overall system.

4.6. Ablation study

To assess the contribution of each component in DualCrossAttnNet, we conduct a series of ablation experiments on the ODIR-2019 dataset. Four model variants are evaluated under identical training settings:

- **v1**: a dual-branch EfficientNet-B2 baseline without cross-attention, gated fusion, SE enhancement, or GeM pooling;
- **v2**: v1 augmented with the SE-based attention enhancement module;
- **v3**: v2 further equipped with the bilateral cross-attention mechanism;

Table 6
Ablation results of the DualCrossAttnNet on the test set.

Model	Test set						
	Accuracy	Precision	Recall	F1	AUC	KS	Score
v1	73.45	71.28	69.15	70.20	81.34	62.18	72.36
v2	78.23	76.45	74.82	75.62	85.67	68.34	77.89
v3	87.92	85.67	83.84	84.75	94.15	82.34	87.57
v4	88.20	91.34	90.62	90.98	98.49	95.05	92.73

Table 7
Ablation results of the DualCrossAttnNet on the validation set.

Model	Validation set						
	Accuracy	Precision	Recall	F1	AUC	KS	Score
v1	75.12	72.89	70.64	71.75	82.67	64.93	74.13
v2	80.15	78.23	76.45	77.33	87.12	70.89	79.67
v3	89.73	87.45	85.92	86.68	95.67	84.92	89.49
v4	85.80	93.67	90.30	91.95	97.87	93.34	91.85

Table 8
Performance comparison on the test set.

Model	Accuracy	Precision	Recall	F1	AUC	KS	Score
BFENet	88.45	89.78	88.63	89.21	91.20	53.50	82.34
Vgg-16	86.12	87.89	86.72	87.30	86.81	44.94	78.35
TL-CCL	87.34	93.58	82.47	87.67	88.92	67.84	83.92
LAD-GCN	85.67	82.34	79.56	80.93	87.45	58.72	79.54
Fundus-DeepNet	89.18	89.88	87.29	88.57	99.76	88.92	91.61
DualCrossAttnNet	88.20	91.34	90.62	90.98	98.49	95.05	92.73

- **v4**: the full DualCrossAttnNet model, which adds the gated fusion module and GeM-based classifier on top of v3.

All variants are trained with the same optimizer, learning rate schedule, data splits, and stopping criteria. The results, summarized in Tables 6 and 7, reveal several key trends. First, the baseline v1 achieves reasonable performance, demonstrating that a dual-branch EfficientNet-B2 backbone already benefits from bilateral information. Adding the SE module (v1→v2) yields a clear improvement in F1, AUC, and composite score, indicating that channel-wise recalibration effectively emphasizes lesion-relevant feature channels and suppresses background noise.

The introduction of cross-attention (v2→v3) produces the largest single-step gain among all components, substantially boosting accuracy and composite score. This confirms that explicitly modeling spatial and channel interactions between the two eyes allows the network to better exploit binocular structural complementarity and pathological correlations. Finally, incorporating the gated fusion mechanism and GeM-based classifier (v3→v4) further improves the composite score and stabilizes performance across metrics by enabling pixel-level, quality-aware fusion and flexible global aggregation. Overall, the composite score increases by approximately 20.37 percentage points from the baseline v1 to the full model v4, highlighting the strong synergy among the proposed components.

4.7. Comparison study

We next compare DualCrossAttnNet with several representative methods from the literature, including classical CNN-based models (e.g., VGG-16 [31]), causal representation learning methods such as TL-CCL [9], graph-based approaches such as LAD-GCN [7], and the state-of-the-art fundus model Fundus-DeepNet [4]. All models are evaluated on the same training/validation/test splits of ODIR-2019 using the metrics described in Section 4.3, and the results are shown in Tables 8 and 9.

Tables 8 and 9 report the performance of all competing models on the test and validation sets, respectively. Among the baselines,

Table 9
Performance comparison on the validation set.

Model	Accuracy	Precision	Recall	F1	AUC	KS	Score	Parameters
BFENet	87.78	89.12	88.09	88.60	90.30	51.30	81.32	25.8M
Vgg-16	86.23	87.78	86.59	87.18	87.05	43.97	78.22	138.4M
TL-CCL	88.67	94.12	83.89	88.71	90.15	69.23	85.16	137.4M
LAD-GCN	86.89	83.78	81.23	82.49	88.67	60.94	81.05	8.05M
Fundus-DeepNet	89.89	89.96	88.31	89.13	99.86	88.92	91.97	28.3M
DualCrossAttnNet	85.80	93.67	90.30	91.95	97.87	93.34	91.85	43.3M

Fundus-DeepNet shows the strongest overall competitiveness, achieving the best test Accuracy (89.18%) and AUC (99.76), as well as the highest validation Accuracy (89.89%), AUC (99.86), and composite score (91.97). TL-CCL attains the highest Precision on both the test set (93.58%) and the validation set (94.12%), but its Recall, F1-score, and composite score remain lower than those of Fundus-DeepNet and DualCrossAttnNet. BFENet provides relatively balanced performance, whereas Vgg-16 yields the weakest overall results despite having the largest model size. LAD-GCN has the smallest parameter count (8.05M), but its F1-score and composite score are comparatively lower, indicating a clear trade-off between model compactness and predictive performance.

Compared with these baselines, DualCrossAttnNet demonstrates a competitive and more recall-oriented performance profile. On the test set, it achieves the best Recall (90.62%), F1-score (90.98%), KS (95.05), and composite score (92.73), while its AUC (98.49) is second only to Fundus-DeepNet. On the validation set, DualCrossAttnNet again attains the best Recall (90.30%), F1-score (91.95%), and KS (93.34), together with a high Precision of 93.67%. However, Fundus-DeepNet still performs better in validation Accuracy, AUC, and composite score, and TL-CCL remains superior in Precision. Therefore, the results suggest that DualCrossAttnNet does not uniformly outperform all baselines on every metric; rather, its main advantage lies in achieving stronger Recall-, F1-, and KS-oriented performance while maintaining a moderate model size (43.3M) relative to larger models such as VGG-16 and TL-CCL.

4.8. Visualization

This section provides a comprehensive visual analysis of the proposed model, including ablation results, confusion matrix analysis, and performance–parameter trade-offs. These visualizations help interpret the behavior of DualCrossAttnNet, reveal the contributions of individual components, and illustrate the balance between accuracy, model complexity, and robustness.

4.8.1. Visualization of ablation results

Fig. 11 presents a heatmap and performance-improvement trend for the ablation experiments (v1–v4). The heatmap clearly shows that the composite score increases monotonically as more components are added, with the full DualCrossAttnNet (v4) achieving a composite score of 92.73%. Compared with the baseline v1, this represents an overall gain of approximately 20.37 percentage points, demonstrating the strong cumulative effect of the proposed modules.

Among all components, the cross-attention mechanism (v2→v3) contributes the largest performance jump, highlighting the importance of explicitly modeling bilateral feature interactions. The SE module and the combination of gated fusion with GeM pooling provide additional but smaller gains, indicating that they primarily refine feature representation and improve the stability of global aggregation. Taken together, these results confirm the positive synergy among the components and validate the overall system performance of the DualCrossAttnNet.

4.8.2. Visualization of model interpretability

To further analyze the decision mechanism of DualCrossAttnNet, we employ Grad-CAM [32] in Fig. 12 to visualize the regions that contribute most to the final predictions. By computing the gradients of the target

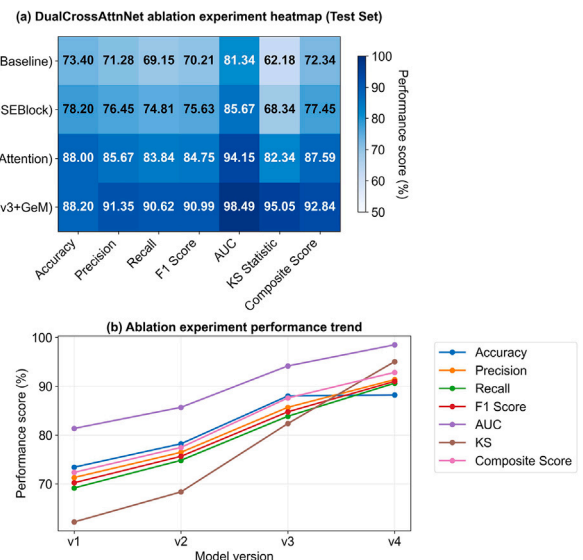


Fig. 11. Ablation results visualization of the DualCrossAttnNet.

category score with respect to the last convolutional feature maps, Grad-CAM produces class activation heatmaps that highlight disease-related retinal regions.

The visualization results show that the high-response regions are mainly concentrated on clinically relevant lesion locations rather than irrelevant background areas. Owing to the dual-branch structure and bilateral cross-attention mechanism, the model exhibits differentiated responses to the left and right eyes, and the activation patterns vary when the severity or manifestation of bilateral lesions is inconsistent. Moreover, different disease categories present distinct spatial heatmap distributions, indicating that the model can adaptively focus on category-related pathological regions. These results support the interpretability and medical plausibility of DualCrossAttnNet, and further enhance its reliability for clinical decision support.

4.8.3. Confusion matrix analysis

Fig. 13 shows the confusion matrix for the eight ocular disease categories on the test set. The diagonal entries are consistently high for major disease classes such as diabetes (D), glaucoma (G), and cataract (C), demonstrating strong recognition capability for clinically common conditions. Off-diagonal entries are generally small, indicating that misclassifications are infrequent. Most residual confusions occur between clinically related or visually similar categories, which is expected in real-world fundus imaging.

The macro-averaged metrics—91.34% precision, 90.62% recall, and approximately 90.5% F1-score—further confirm that DualCrossAttnNet maintains a good balance between sensitivity and specificity across all disease categories. These results, together with the high AUC and KS values, indicate that the model provides reliable multi-label predictions and robust separation between positive and negative cases, which is crucial for practical screening applications.

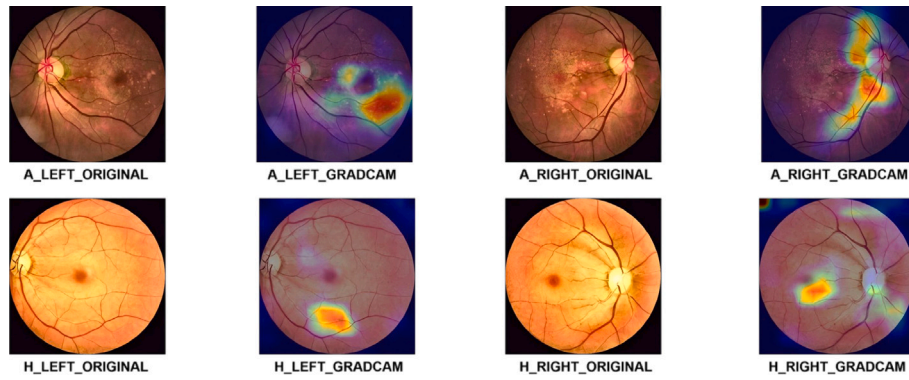


Fig. 12. Grad-CAM based attention visualization of DualCrossAttnNet.

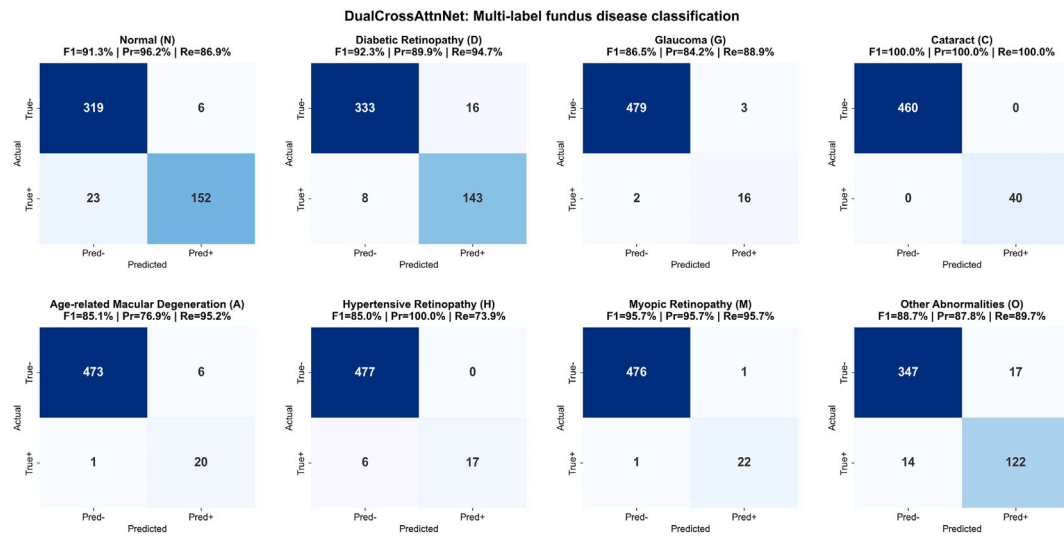


Fig. 13. Confusion matrix visualization of the DualCrossAttnNet on the test set.

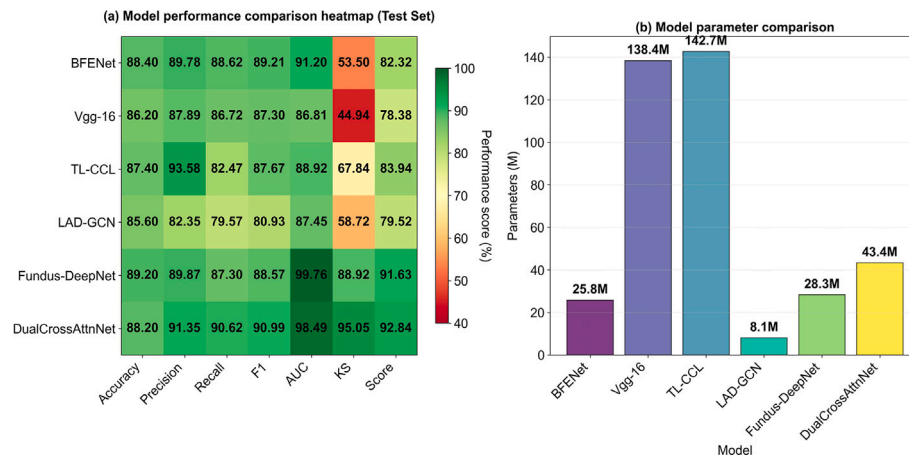


Fig. 14. Visualization of method comparisons. (a) Heatmap-based visualization of the performance differences among methods. (b) Visualization of model parameters for all methods.

4.8.4. Visualization of Comparative Experiments

Fig. 14 visualizes the performance and parameter counts of DualCrossAttnNet and competing methods using a combined heatmap and radar-plot representation. The heatmap highlights that DualCrossAttnNet lies close to the Pareto frontier in the space of

composite score versus parameter count, offering high comprehensive performance without excessive computational demands.

We further examine the performance–parameter trade-off by comparing the positions of different models in this space. DualCrossAttnNet achieves a favorable balance: it outperforms lightweight models such as

LAD-GCN in most metrics while requiring significantly fewer parameters than heavyweight CNNs such as VGG-16. Models that are far from the Pareto frontier either incur high computational costs without commensurate performance gains or sacrifice too much accuracy to reduce model size, indicating suboptimal parameter–performance trade-offs.

The radar plot also shows that DualCrossAttnNet delivers consistently strong results across accuracy, precision, recall, F1, AUC, and KS, whereas other methods often exhibit pronounced weaknesses in one or more dimensions. This comprehensive advantage underscores the effectiveness of the proposed bilateral cross-attention and gated fusion design and highlights the good generalization capability of DualCrossAttnNet across multiple evaluation criteria.

5. Conclusions and future work

In this study, we proposed DualCrossAttnNet, an automated deep learning system for multi-label ocular disease classification based on bilateral fundus images. The framework integrates image preprocessing, bilateral feature extraction, cross-attention-based interaction, gated fusion, and a GeM-based multi-label classifier. Extensive experiments on the ODIR-2019 dataset demonstrate that DualCrossAttnNet achieves a composite score of 92.73%—a 20.37% improvement over the baseline—with test and validation accuracies of 88.20% and 85.80%, respectively. In addition, a five-fold stratified cross-validation study was conducted to further assess model stability under different patient-level data partitions. The results confirm that the reported performance is stable and reproducible rather than dependent on a single train–validation–test split. Ablation studies indicate that the SE attention module, cross-attention, and gated fusion with GeM pooling contribute incremental gains of + 5.53%, +9.68%, and + 5.16% in composite score, with cross-attention providing the largest single-step improvement and underscoring the importance of bilateral feature interaction. With 43.36M parameters, DualCrossAttnNet attains a favorable balance between performance and complexity.

It should be noted that ODIR-2019 provides supervision at the patient/pair level rather than strictly at the single-image level. As a result, bilateral models are more naturally aligned with the annotation protocol, and comparisons with single-image baselines should therefore be interpreted primarily as task-level comparisons under the current dataset setting rather than as purely architecture-only comparisons. Besides, the generalization ability of the proposed framework to other ophthalmic diseases has not yet been fully validated, mainly because ODIR-2019, the benchmark dataset used in this study, covers only a limited spectrum of ocular diseases, while other publicly available datasets are often constrained by small sample sizes and incomplete annotations. Future work will therefore focus on developing more efficient and more generalizable ophthalmic disease diagnosis algorithms for cross-dataset detection of a wider range of ocular diseases.

CRedit authorship contribution statement

Yang Zhou: Writing – review & editing, Methodology, Investigation, Funding acquisition, Conceptualization. **Qiaolu Wang:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization. **Hua Zhong:** Writing – original draft, Methodology, Investigation, Conceptualization. **Gongping Chen:** Writing – review & editing, Validation, Methodology, Conceptualization. **Liqun Xiao:** Writing – review & editing, Validation, Funding acquisition. **Min Wang:** Validation, Methodology.

Ethics approval

This study utilized the publicly available ODIR-2019 dataset. All images in this dataset were collected and released by the dataset providers in accordance with relevant ethical standards and regulations. The present study did not involve any direct interaction with human participants or animals. The project was approved by the relevant ethics committees, and informed consent was obtained from all participants.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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